MATT Improvement Senior Design Project

Meeting Minutes

Meeting Purpose: Understand the current MATT application, identify project goals, clarify technical expectations, and establish key contacts for the senior design team.

Attendees:

- Megan Hopperton — Project stakeholder and consultant
- Samuel Bricking — Project advisor
- Sam Anderson
- Cooper Orlowski
- Luke Link
- Srithan Mopuru

Primary Project: Improve the MATT application to make cyber exercise assessments easier to conduct, improve usability, and generate more useful evaluation results.

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1. Project Background and Purpose

MATT is an application used to evaluate participants during incident response exercises involving opposing Red and Blue teams. The application originated from a United States military tool and is used to track participants' progress as they complete cybersecurity tasks.

The current application has several limitations:

- The interface is not particularly intuitive or visually appealing.
- Accessing and launching the application is cumbersome.
- Recording assessments can be difficult to manage during dynamic and fast-paced cyber exercises.
- Existing results could be refined to make them easier to interpret.
- Assessors need a better way to record specific observations and feedback about participant performance.

Overall goal: Improve MATT's functionality, usability, and presentation while helping assessors produce more detailed and meaningful evaluations.

2. Current MATT Functionality

MATT uses a hierarchical evaluation structure that breaks down cybersecurity activities into increasingly specific tasks.

The evaluation process includes:

- **Core tasks:** Major cybersecurity activities, such as identification and enumeration.
- **Subtasks:** Individual activities within each core task, such as locating a network diagram, mapping the network, confirming the network topology, or identifying who has access.
- **Detailed evaluation criteria:** More granular actions that allow assessors to identify exactly which skills or activities are being evaluated.

The application uses a visual component called the **MATT Wheel** to represent this task structure and track progress.

Assessors can record participant performance using proficiency levels or statuses, including whether a task was attempted, met the standard, or exceeded expectations.

3. Assessment Workflow and User Needs

During cyber exercises, Blue Team members perform prescribed actions based on their objectives and standard operating procedures.

Assessors are embedded with the teams and observe their activities. They use MATT to record progress and evaluate performance.

The workflow involves selecting a core task or subtask and marking it as in progress when the participant begins attempting the activity.

Because exercises can be chaotic and fast-moving, assessors need an interface that allows them to record observations quickly and accurately without disrupting their work.

A key requested improvement is the ability to enter **specific written feedback** alongside performance ratings. For example, an assessor should be able to explain what an individual or team was expected to do, what they failed to accomplish, and what they did instead.

4. Technical Expectations and Development

The team asked whether the project required a particular programming language.

Megan Hopperton indicated that she did not have a specific language requirement.

Samuel Bricking suggested that the team could choose technologies it is comfortable using. Based on his understanding of the existing application, a possible approach could involve:

- A front-end JavaScript framework for the user interface.
- A back-end database for storing assessment information.

The project appears to allow flexibility in the technology stack. However, the team still needs to investigate the current application's architecture and confirm technical requirements before selecting an implementation.

5. Project Contacts and Responsibilities

- **Samuel Bricking:** Project advisor who can guide the team throughout development.
- **Megan Hopperton:** Stakeholder and consultant who can explain the project's goals and assessment requirements.
- **Douglas Bennett:** Expected to be one of the team's primary contacts. He previously served as deputy of assessments and has extensive hands-on experience using MATT.

Douglas Bennett is expected to provide practical insight into the assessment process and the application's shortcomings, although he is not primarily a front-end developer.

6. Discussion Outcomes

The meeting established several key points:

1. The project focuses on improving an existing assessment application rather than simply creating a new interface.
2. Usability is a major priority because assessors must use MATT during fast-paced cyber exercises.
3. The MATT Wheel and hierarchical task structure are important components that the team needs to understand before proposing changes.
4. Written feedback and more refined assessment results are important potential improvements.
5. The team has flexibility in choosing programming languages and development tools, subject to further technical investigation.
6. Douglas Bennett will likely be an important resource for understanding real-world user needs.

7. Action Items

Action Item	Responsible Party	Status
Launch and demonstrate the current MATT instance	Samuel Bricking	In progress during meeting

Explore the existing application and understand its functionality	Senior design team	Pending
Learn how assessors use the MATT Wheel and record evaluations	Senior design team, with stakeholder guidance	Pending
Consult Douglas Bennett about assessment workflows and pain points	Senior design team	Pending
Investigate the existing application's technical architecture	Senior design team	Pending
Identify potential interface, feedback-entry, and reporting improvements	Senior design team	Pending
Determine a suitable technology stack after assessing technical requirements	Senior design team	Pending

8. Questions for Follow-Up

The following questions were not fully answered during the provided portion of the meeting:

- What technologies, frameworks, databases, and infrastructure currently support MATT?
- What are the most common usability problems experienced by assessors?
- Which parts of the current application must be preserved?
- How should written feedback be associated with individual tasks, participants, or teams?
- What types of reports or visualizations would make evaluation results more useful?
- Are there existing user permissions, security requirements, or data-handling restrictions?
- Will the team improve the current application or develop a replacement that preserves its core functionality?
- What are the expected deliverables and criteria for determining project success?

9. Next Steps

The team should gain access to the existing MATT application and document its current workflows before deciding on a technical solution. The next priority is to consult Douglas Bennett and other assessment staff to understand the most significant user problems and establish which improvements would provide the greatest benefit.

Key takeaway: The project's central challenge is to make MATT faster and easier for assessors to use while improving the quality and usefulness of the information collected during cybersecurity exercises.